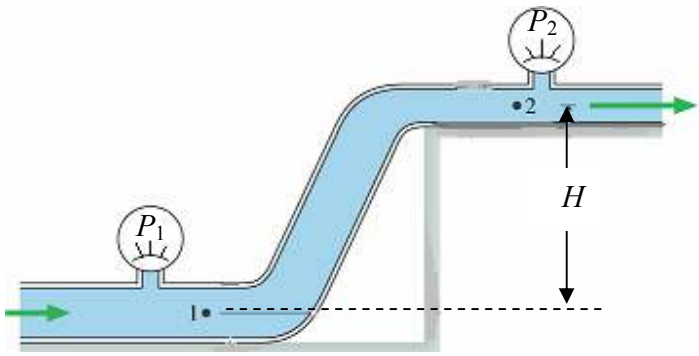
Part A. Multiple Choice.

1. A ball falls to the earth from a height *h* and bounces to a height *h′*. Momentum is conserved in the ball-earth system
   a. no matter what height h′ the ball reaches. *
   b. only if h′ < h.
   c. only if h′ = h.
   d. only if h′ > h.
   e. only if h′ ≥ h.

Solution

An attentive look at the name of the system permits to state that there is no external force in the direction of motion of the ball with respect to the ball-earth system. Therefore, the momentum is conserved as in answer (a).

2. A tap water pipe is bent as shown in the diagram. Sections 1 and 2 of the pipe are horizontal but section 2 is placed higher at a height of *H*. The diameter of the pipe in section 1 is 1.5 times larger than the diameter in section 2. Indicate which of the following is the most complete answer for the relationship between the hydrostatic pressure measured by the gauges in sections 1 and 2:



(a) $P_1 < P_2$ because water flows slower in section 1;
(b) $P_1 > P_2$ because water flows slower in section 1;
(c) $P_1 > P_2$ because section 1 is lower than section 2;
(d) $P_1 > P_2$ because water flows slower in section 1, and section 1 is lower than section 2; *
(e) For some value of H, $P_1 = P_2$ because the lower speed of the flowing water in section 1 is compensated by the hydrostatic pressure due to the difference H in the height of the two sections.

Solution

Bernoulli equation and hydrostatic pressure make the answer (d) correct.

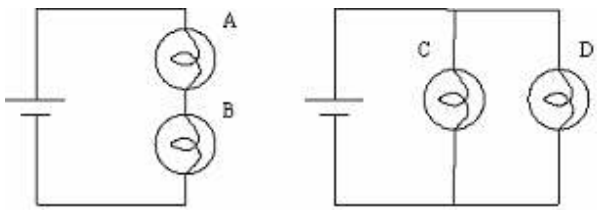
3. We know that a person emits about 500 W of radiation. We also know that a person sitting still uses about 100W of chemical energy. Where is the rest of the energy mainly coming from?

(a) Heat conducted from the air into our body.
(b) Convection of the heat.
(c) Radiation from objects around us. *
(d) The energy, emitted by the body does not have to have a source.
(e) Burning fat, which we accumulate previously.

Solution

Answer (a) is wrong because the temperature of the air is lower. Answer (b) might be correct, but compared to radiation, the convection is less effective. Answer (e) actually refers to chemical energy.

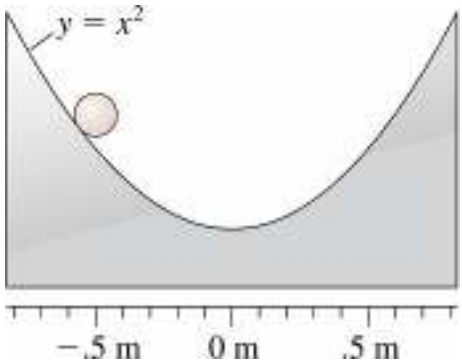
4. In two shown circuits, the two batteries are identical and maintain constant voltage. The bulbs with brightness A, B, C and D, are identical and have resistance R. Assume that the bulbs are brighter when there is more current through them. Which of the following relationships correctly describes the brightness of the bulb?
   (a) A = B > C = D
   (b) C = D > A = B *
   (c) A = B = C = D
   (d) A = C > B = D



Solution

Current through the bulbs A and B is same and is equal to $\varepsilon/(2R)$. The currents through the bulbs C and D are equal to each other and can be obtained as: $\varepsilon/(R/2) = 2\,\varepsilon/R$, which is four times greater than the current in the first circuit The estimation does not consider the internal resistance of the battery.

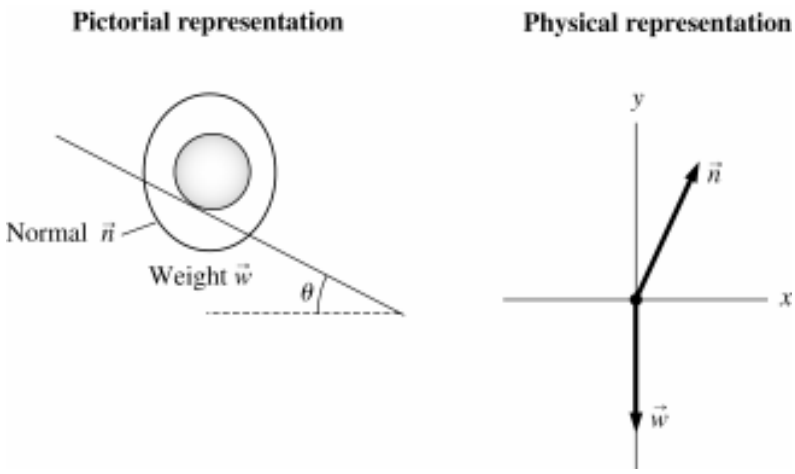
5. The figure on the right shows an accelerometer: a device for measuring the horizontal acceleration of cars and airplanes. The device consists of a ball that is free to roll on a parabolic track. A scale along the bottom is used to measure the ball's horizontal position x. What is the acceleration of the car [in m/s^2] when the displacement of the ball in the accelerometer equals 0.20 m? $g = 9.8$ m/s^2.
   (a) 3.9 * (b) 3.6 (c) 26.5 (d) 24.5 (e) 9.1



Solution

The ball is a particle on a slope, and that the slope increases as the absolute value of x-displacement increases. Assume that there is no friction and that the ball is being accelerated to the right so that it remains at rest on the slope.

Although the ball is on a slope, it is accelerating to the right. Thus we'll use a coordinate system with horizontal and vertical axes.



Newton's second law is

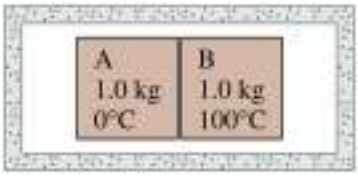
$$\Sigma F_x = n\sin\theta = ma_x \qquad \Sigma F_y = n\cos\theta - w = ma_y = 0 \text{ N}$$

Combining the two equations, we get

$$ma_x = \frac{w}{\cos\theta}\sin\theta = mg\tan\theta \Rightarrow a_x = g\tan\theta$$

The curve is described by $y = Ax^2$, where $A = 1\ \text{m}^{-1}$. Its slope is $\tan\theta$ which is the derivative of the curve. Hence, $\frac{dy}{dx} = \tan\theta = 2Ax \Rightarrow a_x = g(2Ax)$ = 9.8x 0.4= 3.9 m/s^2. **<u>Answer (a) is correct</u>**

6. Objects A and B that are initially separated from each other and well isolated from their surroundings are then brought into thermal contact. Initially T_A= 0C and T_B = 100C. The specific heat of A is less than the specific heat of B. After some time, the system comes to an equilibrium state. The final temperatures are:



(a) $T_A = T_B > 50^oC$; *
(b) $T_A > T_B > 50^oC$;
(c) $T_A = T_B < 50^oC$;
(d) $T_B > T_A > 50^oC$;
(e) $T_A = T_B = 50^oC$.

Solution

Object A has a smaller specific heat and thus less thermal inertia. The temperature of A will change more than the temperature of B.

7. An object is moving at constant speed v_0 towards a source that is at rest and that is emitting sound waves of frequency f_0. The frequency of the echo that returns to the source after being reflected from the object is given by:

(a) $f_{echo} = f_0 \frac{v}{v - v_0}$

(b) $f_{echo} = f_0 \frac{v - v_0}{v + v_0}$

(c) $f_{echo} = f_0 \frac{v + v_0}{v - v_0}$ *

(d) $f_{echo} = f_0 \frac{v + v_0}{v}$

Solution

The frequency is Doppler-shifted to higher values for a detector moving toward the source. The frequency is also shifted to higher values for a source moving towards the detector. The formula is obtained in two steps. First, the object acts like a moving detector and "observes" a frequency that is given by

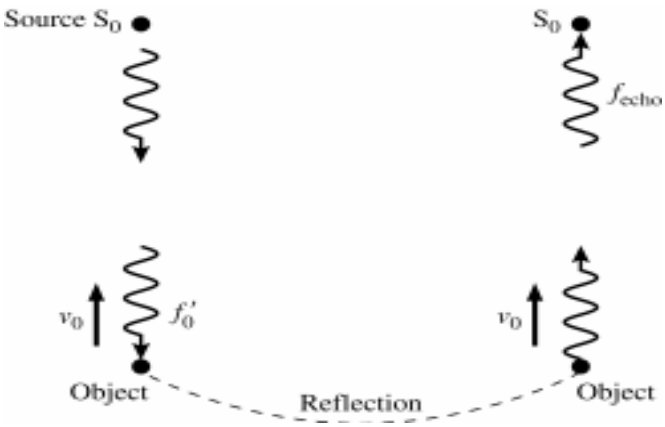
$$f_0' = f_0(1 + v_0 / v) \qquad (1)$$

Second, as this moving object reflects (or acts as a "source" of ultrasound waves), the frequency of echo as observed by the original source S_o is

$$f_{echo} = f_0'(1 - v_0 / v)^{-1} \qquad (2)$$

Combination of equations (1) and (2) gives

$$f_{\text{echo}} = \frac{f_0'}{1 - v_0/v} = \frac{f_0\left(1 + v_0/v\right)}{1 - v_0/v} = \frac{v + v_0}{v - v_0} f_0$$



8. An astronaut lifts off from planet Zuton in a spaceship. The free-fall acceleration on Zuton g_Z is four times less than on the Earth (g). At some moment of liftoff the acceleration of the spaceship is 2.45 m/s^2 (up). The weight of the astronaut at this instant of time W_Z is more than his weight on the surface of the earth W by the factor of

a. 4
b. 2
c. 1
d. 0.5 *
e. 0.25

Solution

The weight on Zuton (W_Z) is a downward normal force that the astronaut exerts on his seat. According to the 3rd Newton's law, the same force is applied to the astronaut from the seat. The equation of motion of the astronaut for the given instant of time is:

$$ma_{up} = W_Z - mg_Z \Rightarrow W_Z = m\left(a_{up} + g_Z\right) = mg\left(\frac{a_{up}}{g} + \frac{g_Z}{g}\right) = W\left(\frac{2.45}{9.8} + \frac{1}{4}\right) = W\left(\frac{1}{4} + \frac{1}{4}\right) = \frac{1}{2}W = 0.5W$$

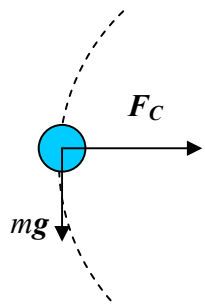
9. A 0.50-kg mass attached to the end of a string swings in a vertical circle with a radius of 2.0 m. When the string is horizontal, the speed of the mass is 8.0 m/s. What is the magnitude of the force of the string on the mass at this position?

a. 16 N *
b. 17 N
c. 21 N
d. 11 N
e. 25 N

Solution

The free-body diagram is sketched in the figure. In the position shown the force of tension is the centripetal force. As there is no horizontal component of velocity at this instant of time, the relationship between the centripetal force F_C and the speed of the object is given by

$$F_C = \frac{mv^2}{R} = \frac{0.50 \cdot (8.0)^2}{2.0} = 16\,\text{N}$$



10. What is the velocity of an electron that passes undeviated through perpendicular electric and magnetic fields if E = 4.0 kV/m and B = 8.0 mT?

a. 32 m/s
b. 500 km/s *
c. $2\text{x}10^{-6}$ m/s
d. 500 m/s
e. 2 km/s

Solution

If the path of the electron in the crossed electric and magnetic fields is linear, as given, the motion of the electron can be only uniform, i.e. to take place without acceleration and hence,with zero net force.

$$eE = evB;\quad v = E/B;\quad v = \frac{E}{B} = \frac{(4.0\times10^{3})}{(8.0\times10^{-3})} = 5.0\times10^{5}\,\frac{m}{s} = 500\,\frac{km}{s}$$

11. A body is oscillating harmonically in a vertical direction being suspended from two identical springs connected in series. The frequency of oscillation is f_1. After the springs are disconnected and attached to the body in parallel, the frequency of the vertical oscillation of the body is equal to

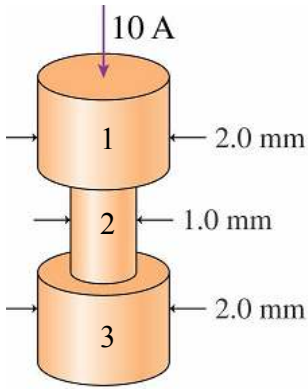
a. f_1.
b. $2f_1$. *
c. f_1 /4
d. $4f_1$.
e. $f1/2$.

Solution

When the springs are connected in series, their elongations are equal and the total elongation is two times greater than each of them. In equilibrium $mg = k\Delta_1 = k\Delta_2 = k_{eff}\,2\Delta$ The frequency is calculated as : $f_1 = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m}} = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$. When the springs are connected in parallel, $mg = k\Delta_1 + k\Delta_2 = k_{eff}\,\Delta = 2k\Delta$

$$f_2 = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m}} = \frac{1}{2\pi}\sqrt{\frac{2k}{m}} = 2f_1$$

12. A piece of aluminum wire shown on the drawing, is connected to a circuit with a source of constant current. The current flowing into segment 1 is I_1 = 10 A. Choose the correct conclusion on the values of current, amount of heat P emitted per second by a unit of length and electric field strength E inside the segments of the wire



a. $I_1 = I_2 = I_3$; $P_1 = P_2 = P_3$; $E_1 = E_2 = E_3$.
b. $I_1 = I_3 = ½\, I_2$; $P_1 = P_3 = 1/16\, P_2$; $E_1 = E_2 = E_3$.
c. $I_1 = I_3 = ½\, I_2$; $P_1 = P_3 = 1/16\, P_2$; $E_1 = E_3 = ½\, E_2$.
d. $I_1 = I_2 = I_3$; $P_1 = P_3 = ¼\, P_2$; $E_1 = E_2 = E_3$.
e. $I_1 = I_2 = I_3$; $P_1 = P_3 = ¼\, P_2$; $E_1 = E_3 = ¼\, E_2$. *

Solution

Segments 1 and 3 are identical from any point of view.
Current is everywhere same (law of conservation of electric charge).
The power, emitted by the unit of length is inversely proportional to the cross-sectionsl area of the wire; therefore, P_2 must be four times as great as P_1 or P_3.

Electric field lines must be confined within the wire. The number of lines penetrating the unit of surface area perpendicular to the lines is proportional to the value of E. The density of the electric field lines inside the segment 2 is four times greater than the densit of the lines in segments 1 or 3. So, $E_2 = 4E_1$.

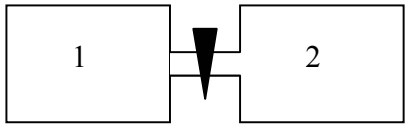
13. A spacecraft (mass = m) orbits a planet (mass = M) in a circular orbit (radius = R). What is the minimum energy required to send this spacecraft to a distant point in space where the gravitational force on the spacecraft by the planet is negligible?

a. $GmM/(4R)$
b. GmM/R
c. $GmM/(2R)$ *
d. $GmM/(3R)$
e. $2GmM/(5R)$

Solution

A spacecraft on a circular orbit of radius R has potential energy of $-GmM/R$ and kinetic energy of $GmM/2R$. So, the total mechanical energy of the spacecraft is equal to the sum of the potential and kinetic energies, i.e. $-GmM/2R$. The minimum work required to move the spacecraft from this orbit to the infinitely far place in the space is equal to the change in the energy of the object from the initial value to zero, which gives answer (c).

14. Two identical thermally isolated containers are separated by a valve. Initially, there is an ideal gas in container 1, and there is a vacuum in container 2. Some time after the valve is opened, the gas in the two containers comes to an equilibrium state. Which of the following statements about the gas in this process is true:



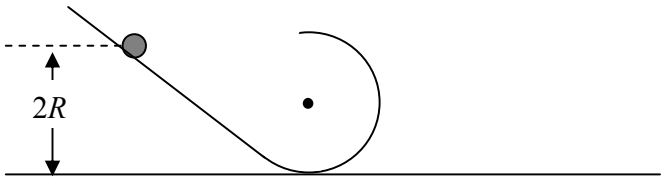
(a) The molar mass of the gas decreases;
(b) The work produced by the gas is zero; *
(c) The temperature of the gas drops;
(d) The work produced by the gas is positive and is equal to the absolute value of the change of the internal energy of the gas.

Solution

The molar mass cannot change if the gas is same. In this problem, the expansion into vacuum is adiabatic. The expansion into vacuum produces no work (no external pressure to produce the work against to). Therefore, answer (b) is correct. (c) cannot take place, because the internal energy of the gas does not change.

15. During the winter vacation children use snow and water to produce frictionless hillsides of different shapes for freestyle exercises. The side view of one of them is shown in the figure: a linear segment of the path is smoothly transferred into a piece of circumference with radius R. A puck starts sliding down from rest at initial height of $2R$. The acceleration of the puck in the lowest point of its trajectory is

(a) g.
(b) $2g$.
(c) $3g$.
(d) $4g$.*
(e) 0.



Solution

Law of conservation of energy gives: $mg2R = ½\, mv^2$; $\rightarrow$ $v^2 = 4Rg$.

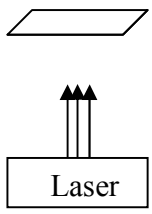
Centripetal force in the lowest point of the trajectory is the net force; hence, the acceleration of the puck is equal to $v^2/R = 4g$.

16. A star undergoes a supernova explosion. Just after the explosion, the material left behind forms a uniform sphere of radius 8.0×10^6 m with a rotation period of 15 hours. This remaining material eventually collapses into a neutron star of radius 4.0 km with a period of rotation T of:
(a) 14 s
(b) 3.8 h
(c) 0.021 s
(d) 0.014 s *
(e) 0.0075 h

Solution

The solution is based on the equation $I_1\omega_1 = I_2\omega_2$. Regardless the shape of the rigid object, its moment of inertia I is proportional to the product mR^2. In this problem, the mass of the object is constant, therefore, $\omega_2 = \omega_1\left(\frac{R_1}{R_2}\right)^2$. The period of rotation is $T = \left(\frac{4.0\times10^3 m}{8.0\times10^6 m}\right)^2 (15h) = 3.75\times10^{-6} h = 0.014$ s.

17. An impulse laser may be treated as a source of photons that are emitted during the time interval of the pulse which is followed by a time interval when no photons are produced. Pulses are periodically repeated. The laser beam of diameter d = 10 microns is directed upward and is perpendicular to a surface of a thin foil with an index of reflection ρ = 0.50 (see the sketch of the experiment). The index of reflection of the surface is the ratio of the reflected energy to the impact energy. A pulse with duration of 0.13 ms has a total energy of 10 J. What must be the mass of the piece of foil that may be supported in the air solely by the light pressure of the laser beam?



(a) < 39 g.
(b) < 3.1×10^{-12} g.
(c) < 39 mg. *
(d) < 3.7 g.
(e) < 0.38 g.

Solution

The given value of the diameter of the beam is not used in the solution. Upon the definition of force, it is the rate of change of linear momentum. The linear momentum changes by its initial value $p_{initial}$ (total absorption) or by the product of $p_{initial}$ and $[1 + \rho]$ (reflection with coefficient ρ).

For a photon, the linear momentum $p = \frac{E}{c}$ (E is the energy of the photon, c is the speed of light). When the laser beam supports the piece of foil, the weight of the foil is equal to the normal force of light pressure. Summary:

$$mg = \frac{\Delta p}{\Delta t} = \frac{p_{initial}(1+\rho)}{\Delta t} = \frac{E}{c}\frac{1+\rho}{\Delta t}; \quad m = \frac{1}{g}\frac{E}{c}\frac{1+\rho}{\Delta t} = \frac{10\times1.5}{9.8\times3\times10^8\times0.13\times10^{-3}} = 3.92\times10^{-5}\text{ kg}$$

= 39 mg

18. Two sinusoidal waves traveling at the same speed in opposite directions interfere to produce a standing wave with the wave function y = (1.50 m) sin(0.400x) cos(200t), where x is in meters and t is in seconds. The speed of propagation of each of the interfering waves is
(a) 159 m/s
(b) 200 m/s

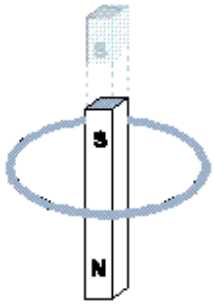
(c) 300 m/s
(d) 47.7 m/s
(e) 500 m/s *

Solution

$y = (1.50\text{ m})\sin(0.400x)\cos(200t) = 2A\sin(kx)\cos(\omega t)$.
Therefore, $k = 0.400$ rad/m; $\omega = 200$ rad/s. The speed of the traveling waves in the medium is

$$v = \lambda f = \frac{\lambda}{2\pi}2\pi f = \frac{\omega}{k} = \frac{200}{0.400} = 500\frac{m}{s}$$

19. A bar magnet is dropped from above and falls through the loop of a wire as shown. A student measures the current in the loop between some instant when the north pole of the magnet is above the plane of the loop and the other instant when the south pole of the magnet is below the plane of the loop.
Which statement is correct about the result of the student's measurement?



a. The current in the loop flows in one direction increasing steadily to its maximum value when the center of the bar crosses the centre of the loop plane, after which the current begins to steadily decrease.
b. The current in the loop is harmonically oscillating because the magnetic flux through the loop is changing.
c. The current in the loop flows first in one direction, then, after the center of the bar crosses the centre of the loop plane, the current begins to flow in the opposite direction. *
e. No current flows in the loop because both ends of the magnet move through the loop.

Solution

The magnetic field lines are exit from the north pole of the bar magnet and enter the south pole of the magnet. Inside the bar magnet, the magnetic field lines are directed from the south pole to the north pole. When the number of the filed lines inside the wire loop is increasing (the magnetic flux is increasing), the induction produces a current of the direction that creates a magnetic field of the opposite direction. As the initial direction of the magnetic field lines is downwards, the current of induction produces the magnetic field directed upwards, so the current must flow counterclockwise. After the center of the magnet passes the plane of the loop, the number of lines, penetrating the loop, begins to decrease. According to the Lenz's law, the current of induction switches to the opposite direction to create a magnetic field parallel to the decreasing external magnetic field.

20. Unpolarized light is passed through three successive Polaroid filters, each with its transmission axis at 45° relative to the preceding filter. What percentage of the light gets through?

a. 0%
b. 12.5% *
c. 25%
d. 50%
e. 33%

Solution

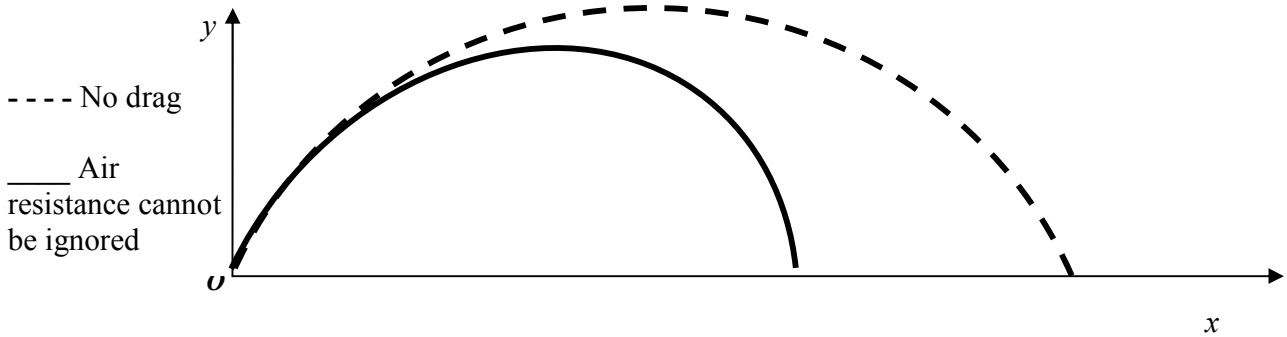
After the first filter, the intensity is $I_1 = ½ I_0$; after the second filter the passed intensity becomes $I_2 = ½ I_0 \cos^2(45^o) = ¼ I_0$; after the third filter, the intensity is $I_3 = I_2 \cos^2(45^o) = 1/8\ I_0 = 0.125\ I_0$.

Part B.

1. The drag force due to air resistance on a falling object depends on the instantaneous velocity of the object as $D = -1/4\ Av^2$. Sketch a diagram of height vs. distance for the trajectory of two projectile objects launched from the same point at ground level with the same angle to the horizontal: a) – the projectile for which the air resistance is negligible, and b) - the projectile that is experiencing a drag force in the air.

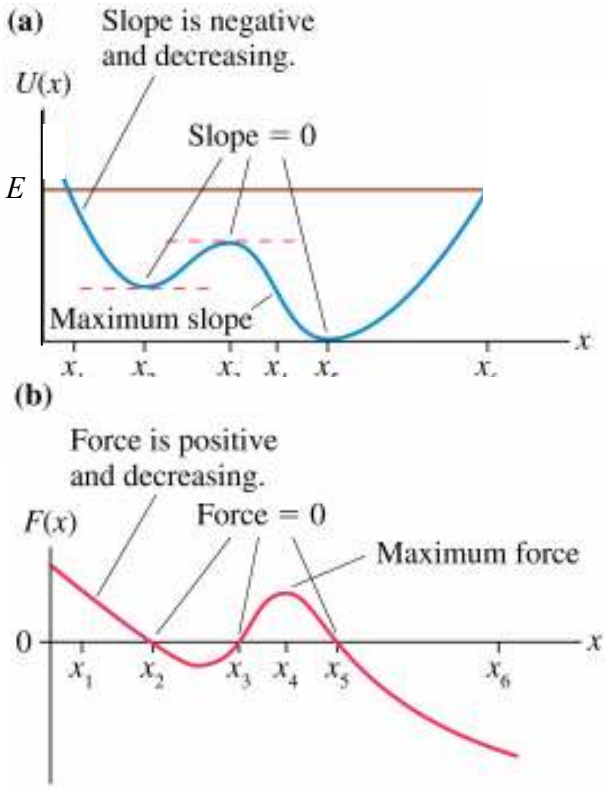
Solution

The drag will affect the *x*- and the *y*-component of the velocity of the projectile everywhere through its path. So, the lower height and the shorter horizontal distance are expected to be observed for an object that experiences the drag force due to the air resistance. The shape of the trajectory must change correspondingly.



2. The Figure below shows a diagram of the potential energy of an object in an isolated mechanical system with the total energy E and conservative forces only. On the space below the given diagram, sketch a diagram for the *x*-component of the net force on the object as a function of x. Your diagram must show the correct trend of the function, zeros and vertex points.

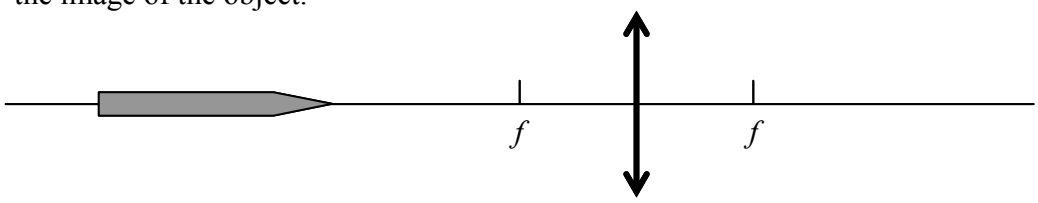
Solution



To draw a diagram, it is helpful to refer to some simple example of motion of a point mass in the field with maximum or minimum of the potential energy function. The simplest example is the motion of a point mass along a hill with a profile of the same shape as the shape of the given function of the potential energy. In this case, the potential energy of the object is the energy in the gravitational field of the earth, and the surface of the path is supposed to be frictionless. For this approach:

1) If the object is carefully placed in either of three points with zero slope of the potential energy function, it will remain at rest for the infinite time; so, it is reasonable to conclude that the net force is zero at the equilibrium positions; we expect either a maximum or a minimum value of the x-component of the force to appear between two zeros.
2) If the object is gently placed on the sections of the hill with positive slope, it will start moving in the negative x-direction; so, these sections must have negative values of force.
3) If the object is placed on the steep section of the hill, the acceleration would be greater in its absolute value; so, the steeper sections of the function of the potential energy must correspond to the greater magnitude of the force.

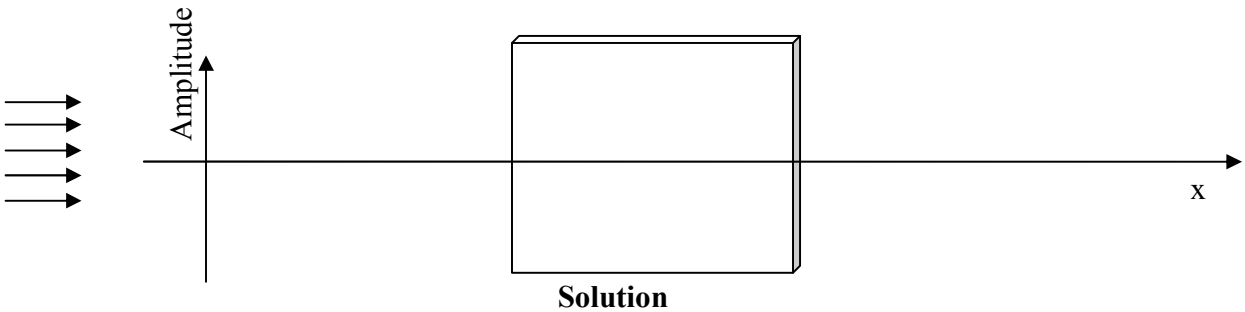
3. A very thin stick is placed on the optical axis of a thin convex lens as shown in the diagram. Draw the image of the object.



Solution

As a possible method of solving the problem, one can draw the vertical objects at the ends of the stick and then, construct the two images of these auxiliary objects to find the intersections of the images with the optical axis of the lens.

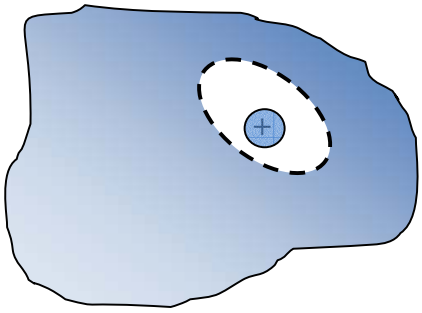
4. The monochromatic light is produced by a laser and propagates through vacuum from left to right until it strikes normally the surface of the glass brick, as shown. The direction of the wave propagation coincides with x-axis.
On the system of coordinates "amplitude vs. position", sketch the wave before it enters the glass brick, inside the glass brick and after it exits the brick.



Solution

The students are expected to draw a sinusoidal function with the wavelength shorter inside the brick, because of decrease in the speed of propagation in the medium with index of refraction >1. To the left and to the right from the brick, the wavelength must be same. However, the full mark must be given to those who will also show the decrease in the amplitude inside the brick and the smaller amplitude to the right from the brick due to the absorption in the glass.

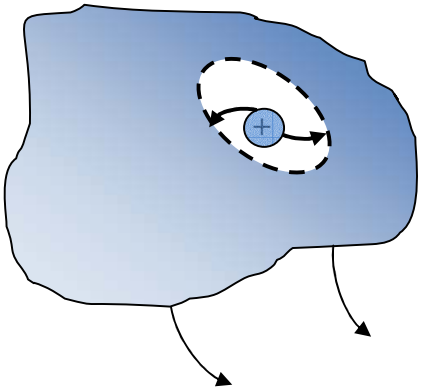
5. An object, shown in the figure, is made out of a conductive material and has a hole completely hidden inside its body. A point charge $+Q$ is kept motionless in the hole by some external force. Sketch the electric field lines inside the hole, inside the conductor and outside the conductor as close to reality as it is necessary to be consistent with the laws of electromagnetism.



Solution

Electric induction causes accumulation of negative charges on the inner surface of the hole with the total negative charge of $-Q$. The electric field lines go from the point charge into the surface of the hole and enter the surface of the hole normally. There must be no electric field lines inside the body of the conductor.

Due to the redistribution of electric charge in the conductor, the outer surface of the object becomes positively charged with the total charge of $+Q$. The electric field lines exit the outer surface of the object under the right angle everywhere and are directed into infinity.



Part C.

Problem 1.

An ambulance needs to be delivered to a remote town devastated by a major earthquake. All roads leading into the town are blocked due to the earthquake and the ambulance can only be rushed to the area by airlift. The ambulance will be pushed out of a military cargo jet at 3000 m altitude and rescue staff needs to find out what kind of parachute is needed for this mission.

The drag force is given by the approximate formula:

$F = \frac{1}{4}\rho A v^2$, where ρ is the density of air and $\rho = 1.2$ kg/m^3.

What should be the diameter of the parachute be so that the ambulance can land softly/safely?

A is the area of the cross-section of the parachute perpendicular to the motion and v is the velocity. Notice that you have to figure out what is approximately the safe landing velocity and what is a weight of a typical ambulance.

Solution

The estimated mass of the ambulance is 3000 kg; and the estimated landing velocity is 5 m/s.

Acceptable assumptions:

v will be anything from 1 to 10 m/s

mass of the ambulance - from 1000 to 10 000 kg

Drag force must be equal to force of gravity at given velocity to ensure constant velocity:

$m\,g = \frac{1}{4}\rho A v^2$

$A = 4mg/\rho v^2 = 4\cdot 3000\text{kg}\cdot 10\text{m/s}^2/(1.3\ \text{kg/m}^3\cdot 25\text{m}^2/\text{s}^2) = 3700\text{m}^2$

Which results in a diameter value of 68 m

Still give the points when they end with the area of the parachute and don't show the radius or diameter.

Problem 2.
A very thin beam of protons is injected at non-relativistic velocities in a circular particle accelerator of radius R. The mass m and the charge e of the proton are well known. The initial current in the accelerator is I and the total number of particles is n. The magnetic flux through the beam circuit changes at a rate of ρ Wb/s, while the radius of the beam track remains unaltered.

What is the value of the current after one turn of the particles?

Solution

To simplify the problem let us assume that the bunch of protons is injected and then circulate in the accelerator chamber as a sole particle with charge ne. Upon the definition a current is a rate at which electric charge flows through the cross section of conductor. If the motion of protons was uniform the relationship between the value of current I_U and speed v_U in the chamber could be expressed as:

$$I_U = \frac{en}{T} = \frac{env_U}{2\pi R}$$

Factor $\frac{en}{2\pi R}$ does not depend on time. That is why just the same formula can be used for the instantaneous current $I(t)$ and speed $v(t)$ of protons:

$$I(t) = \frac{en}{T} = \frac{env(t)}{2\pi R} \qquad (1)$$

To calculate current after one turn it is necessary to determine the speed of protons after one turn. It can be done using the law of conservation of energy and the law of electromagnetic induction. The magnitude of electromotive force of induction is equal to φ, its work produced to accelerate a proton is $e\varphi$. This work results in increasing of the kinetic energy of a proton. After one turn a speed of a proton can be calculated from the law of conservation of energy as:

$$v_1 = \sqrt{v_0^2 + \frac{2e\varphi}{m}} \qquad (2)$$

where v_0 is initial speed of a proton, and can be obtained from Eq.(1) with given initial current I:

$$v_0 = \frac{2\pi RI}{en} \qquad (3)$$

Combining Eqs.(1), (2), and (3) we can calculate current I_1 after one turn as:

$$I_1 = \frac{en}{2\pi R} v_1 = \frac{en}{2\pi R}\sqrt{\left(\frac{2\pi RI}{en}\right)^2 + \frac{2e\varphi}{m}} = I\sqrt{1 + \frac{n^2 e^3 \varphi}{2\pi^2 R^2 m I^2}}$$

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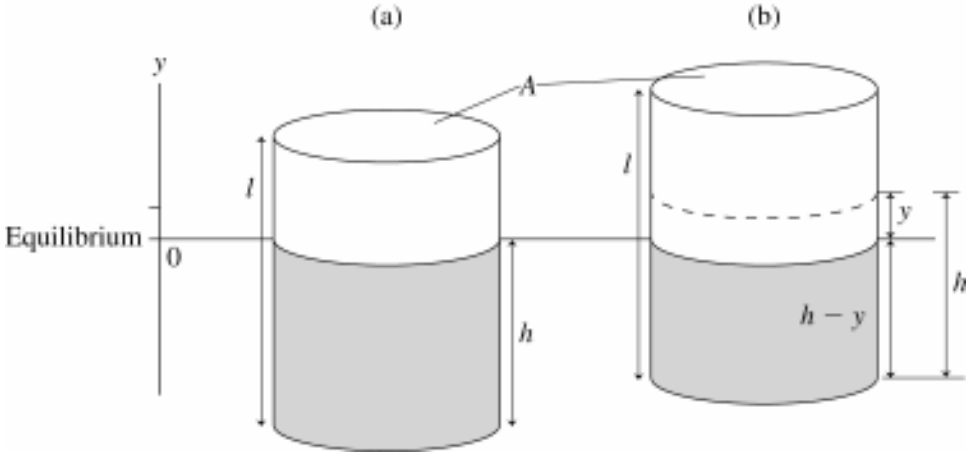
Problem 3.
Very cold fresh water fills a vessel with a depth of 3.00 m and a diameter of 1.00 m. A vertically maintained ice cylinder with diameter d =30.0 cm and length l = 30.0 cm is carefully moved downwards and submerged in the water perpendicular to its surface and in the centre of the surface of the vessel. When ¾ of the length of the cylinder is submerged, it is released. The density of the ice is ρ_i = 917 kg/m^3; the density of the water is ρ_w = 1.00x10^3 kg/m^3.

1) Describe the behavior of the cylinder after it is released.

2) Ignoring the melting of the cylinder during the first few seconds after it has been released, calculate the position of the center of mass of the cylinder in equilibrium, taking the level of the water surface as zero and the vertical axis aimed upwards.
3) Write and solve the equation of motion for the cylinder valid during the first few seconds after it is released. The equation must have a solution in the form *y(t)* where *y* is the displacement; and *t* is time.
4) Explain the physical significance of all parameters of the function *y(t)* and give the numerical values for the parameters.

Solution

The diameter of the cylinder plays no role in the phenomenon.
The buoyant force on the cylinder is given by Archimedes' principle.



1) Because the density of the ice is less than the density of the water, the center of mass of the cylinder must be under the surface in equilibrium. Just after the cylinder is released, it starts to move downwards and will pass its equilibrium position due to inertia. When the center of mass of the cylinder appears below its position of equilibrium, the buoyant force being directed against the displacement of the center of mass starts to push the cylinder up. This results in oscillatory motion of the cylinder.

2) The equilibrium position of the center of mass gives for the displacement Δy of the center of mass the following relationship: $\rho_i\, l = \rho_w\,(l/2 + \Delta y) \rightarrow \Delta y = l\left(\frac{\rho_i}{\rho_w} - \frac{1}{2}\right) = 0.3\left(\frac{917}{1000} - 0.5\right)m = 0.125$ m = 12.5 cm

The position of the center of mass of the cylinder in equilibrium is y = -12.5 cm. It is 5 cm lower than the position of the cylinder at the instant it was released.

3) A is cross-sectional area of the cylinder. Variables y and h are shown in the figure.
In static equilibrium, and we can write $h = (\rho_i/\rho_w)\, l$ (1)
and h = (15 + 12.5) cm = 27.5 cm
The volume of the displaced liquid is $A(h - y)$. Applying Newton's second law in the y-direction,
$\sum F_y = -\rho_i\, lgA + \rho_w\,(h - y)gA$
Substitution of result (1) gives $\sum F_y = -\rho_w\, ygA = a_y m$ (2)
The result in (2) is $F = -ky$, where $k = \rho_w\, gA$. This is Hooke's law, and equation (2) is the equation of a simple harmonic motion.

The angular frequency for the resulting simple harmonic motion is $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{\rho_w gA}{m}} = \sqrt{\frac{\rho_w gA}{\rho_i lA}} = \sqrt{\frac{g}{h}}$,

and the period is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{h}{g}} = 1.10$ s

$y(t) = (12.5 - 7.5)\cos\left(\sqrt{\frac{g}{h}}t\right)cm = 5\cos 5.97t$ cm = $5.00\text{x}10^{-2}\cos(5.97t)$ m